

AuraFlow: Privacy-First NPU-Driven Desktop Automation and Task Synthesizer

A Proposal for the Snapdragon® AI Lab Build & Present Challenge

Target Hardware: Snapdragon®-powered HP PCs	Core Optimization: Qualcomm® AI Hub Models
Primary Category: Productivity & Workflow Automation	Deployment Strategy: On-Device NPU (Zero-Cloud)

1. Executive Summary

AuraFlow is an innovative, next-generation desktop productivity suite engineered exclusively to harness the hardware-accelerated computational capabilities of Snapdragon-powered HP PCs. Operating entirely local (on-device), AuraFlow serves as a context-aware task synthesizer, cross-application orchestrator, and intelligent workflow automator. By leveraging optimized small language models (SLMs) and audio processing models deployed directly onto the system's Neural Processing Unit (NPU) via the Qualcomm AI Hub, AuraFlow delivers lightning-fast, zero-cloud desktop assistance without compromising user data privacy or draining system battery life.

2. Key Capabilities & Innovations

- **Intelligent Action-Item Synthesis:** Local audio streams from ecosystem meetings or native mic input are processed in real-time to generate structured summaries, outstanding tasks, and action timelines seamlessly.
- **Cross-Application Task Orchestration:** Employs an on-device contextual AI router capable of understanding multi-step natural language commands (e.g., 'Draft a follow-up email to the engineering lead based on my last meeting summary and organize the files in my review folder') and executing them locally across application layers.
- **Zero-Cloud Privacy Paradigm:** Operates completely offline. Sensitive company intellectual property, email drafts, conversational content, and business documents never leave the local environment, satisfying strict modern compliance regimes.
- **Dynamic Context Retrieval:** Interacts with a lightweight, NPU-accelerated vector index embedded locally, allowing instantaneous semantic search across open windows, recent files, and user notes.

3. Technical Implementation & Snapdragon Architecture Optimization

To maximize technical implementation scores under the challenge guidelines, AuraFlow utilizes specialized neural architectures tailored for the Snapdragon NPU compute engines:

- **Qualcomm AI Hub Optimization:** Employs quantised text-generation models (such as Llama-3-8B-Instruct or Phi-3-Mini INT4) and whisper-variant audio transcription models carefully compiled for target Snapdragon SoC runtimes.
- **Hardware-Accelerated Execution:** Utilizes ONNX Runtime or Qualcomm Neural Processing SDK to delegate inference layers directly to the system NPU. This architecture completely bypasses traditional cloud latencies and guarantees massive energy savings compared to standard CPU or discrete GPU inference paradigms.
- **Responsive Inter-Process Communication:** Bridges model generation outputs directly with native Windows OS APIs via a highly optimized C++/Python application interface layer, facilitating instantaneous file actions and application macros.

4. Alignment with Hackathon Evaluation Criteria

Evaluation Criterion	AuraFlow Strategic Alignment
Technical Implementation	Directly leverages ONNX/Qualcomm runtimes to offload multi-modal workloads (Speech-to-Text, LLMs) to the Snapdragon NPU, achieving maximum hardware utility.
Application Use Case & Innovation	Solves the critical paradox of enterprise productivity by offering deeply integrated, contextual workflow intelligence while preserving absolute data sovereignty.
Deployment & Accessibility	Packaged as a lightweight, low-footprint background asset purpose-built for seamless setup on premium HP PC architectures running on-device inference.
Presentation & Documentation	Accompanied by a transparent architecture overview, comprehensive benchmark profiles demonstrating NPU speed advantages, and step-by-step modular configuration guides.